Experiment 3
Economic Evaluation Report of Software Engineering Project

Course Project: Embedded Cross-Course AI Agent General Architecture Platform

1. Objective of the Experiment

This experiment takes the current project as the object of analysis and focuses on the economic evaluation of a software project. The core task is to establish a complete evaluation framework from the perspectives of application scenarios, business objectives, external environment, cost inputs, revenue sources, and funding arrangements, and then prepare relevant financial tables to calculate indicators such as net present value, internal rate of return, and dynamic payback period, so as to judge whether the project is economically feasible.

Through this experiment, the goal is to master the basic logic of software project economic evaluation, understand the relationship among software pricing, size estimation, and cost estimation, become familiar with the organization of supporting financial statements, and improve the ability to transform engineering parameters into economic analysis parameters.

2. Experimental Principle

The economic evaluation of a software project is essentially a systematic analysis of how much is invested, how much is recovered, and when the recovery occurs within a given period. The evaluation requires estimating construction investment, working capital, operating cost, operating revenue, and financing conditions under the current price system, tax rules, and industry benchmark parameters, and then organizing them into cash flow data.

This experiment mainly adopts the cash flow analysis method. The pre-financing analysis focuses on the investment return capability of the project as a whole, so the project investment cash flow statement is used. The post-financing analysis examines the return change from the perspective of equity capital after introducing external funding support, so the equity capital cash flow statement is used. The evaluation mainly uses three indicators: net present value (NPV), internal rate of return (IRR), and dynamic payback period (DPP). When NPV is positive, IRR is higher than the benchmark rate of return, and DPP is within an acceptable range, the project can be considered economically feasible under the current assumptions.

3. Experimental Content and Process

3.1 Collection of Basic Information

3.1.1 Business Objective of the Project

The course project is named 'Embedded Cross-Course AI Agent General Architecture Platform', and its main practical scenario is the construction of smart courses in universities. The project aims to provide reusable and extensible AI-assisted teaching services for different courses through unified platform capabilities, thereby reducing the repetitive investment caused by building intelligent functions for each course separately.

The project intends to address several issues, including fragmented construction of intelligent teaching capabilities across courses, inconsistent platform interfaces and technical implementation, difficulty in reusing high-quality AI capabilities at scale, and the lack of closed-loop coordination among question answering, grading, exercise generation, and learning analytics. Based on this, the business objective of the project can be summarized as building a course-level AI teaching platform that can serve multiple courses, provide teachers and students with functions such as question answering, material retrieval, assignment grading, personalized exercises, and learning analytics, and generate economic value through annual course-level service fees, customized configuration service fees, and follow-up support.

3.1.2 Market, Economic, and Social Environment Analysis

Before conducting the economic evaluation, it is necessary to determine whether the external environment supports the implementation of the project. Public information shows that educational digitalization, smart education platform construction, and platform-based course resources have become clear development trends. Resource integration, process datafication, and intelligent services in education already have a realistic foundation. The continuous development of the National Smart Education Public Service Platform further indicates that platformization and digital transformation in education are being steadily promoted.

From the perspective of technology and policy, the application of artificial intelligence, especially generative AI, in education has gradually entered the stage of policy guidance and practical exploration. UNESCO documents on AI and education governance point out that AI can support educational governance, teaching improvement, and learning assistance, and has potential in personalized learning, automated feedback, and educational support. At the same time, issues such as data protection, transparency, teacher training, and responsible use must also be considered.

From the demand perspective, the use of generative AI in educational scenarios has become an expanding direction in both research and practice. Relevant studies generally indicate that such technologies are valuable in personalized learning support, automatic feedback, content generation, tutoring, and the evaluation of complex tasks. Meanwhile, studies on teachers show that university educators' awareness and acceptance of generative AI tools are increasing, which suggests that this kind of project has a realistic basis for adoption.

From the economic perspective, universities usually focus on initial construction cost, follow-up maintenance cost, and the reusability of platform capabilities when purchasing or building educational systems. Compared with the approach of developing an independent system for each course, this project emphasizes unified capability reuse, which in theory helps reduce the average construction cost per course and improve the long-term input-output ratio. Therefore, from the perspective of the external environment, the application scenario targeted by this project is valid, but its economic viability still needs to be verified through subsequent cost and revenue analysis.

3.1.3 Techno-Economic Constraints Analysis

Although the project has a clear application direction, there are still several techno-economic constraints that need to be considered separately in the evaluation. First, the initial development investment is relatively high. The platform includes not only front-end, back-end, and database modules, but also AI agent integration, course knowledge retrieval, platform embedding, and test validation, which require substantial human and development resources in the early stage. Second, the project incurs continuous operating costs after deployment, including server expenses, large-model API calls, and routine maintenance, all of which directly affect the economic performance in the operating stage.

In addition, the current project still has limited validation with real teaching platforms and large-scale concurrent scenarios. This means that further investment will still be required in stability optimization, interface compatibility, and exception handling during actual promotion. If the scope of application expands, content management, access control, quality evaluation, and teacher training across different courses will also increase the governance difficulty of the platform and bring additional organizational and management costs. Therefore, the economic evaluation of this project should not focus only on one-time development investment, but also on operation and maintenance, model invocation, and possible optimization, governance, and training costs in the expansion stage. Among these, the first two categories have been quantified in the tables of this report, while later optimization and governance-training costs are explained as potential extended costs and are not listed separately.

3.1.4 Business Model and Pricing Strategy

This project is suitable for a B2B/B2E service model oriented toward universities, schools, or course teams. Considering the current product form, a platform-hosted service mode is more appropriate, that is, the platform side operates and maintains the system uniformly, while course teams use the related AI teaching services on a course basis.

In terms of pricing, the project is not suitable for directly applying the traditional software license charging model, nor is it exactly the same as institution-level procurement of a full teaching platform. Therefore, a combination of continuing service fees and service fees for implementation-related work is more appropriate. The annual course-level service fee mainly covers long-term services such as platform operation, question answering support, assignment grading, learning analytics, and exercise generation. The customized configuration service fee mainly corresponds to customized configuration and adjustment-optimization related services. Since comparable domestic education platforms and teaching tools more often adopt institution-level cooperation, institutional procurement, or customized quotation models, publicly available per-course prices are limited. Therefore, this report uses 'cost estimation + pilot-stage acceptability' as the pricing basis. After considering the scope of project capabilities, the cost of continuous platform service, and the level acceptable in the pilot stage, the annual course-level service fee is set at RMB 10,000 per course per year, and the customized configuration service fee is set at RMB 12,000 per case for the subsequent economic evaluation.

3.1.5 Project Scale, Cost Estimation, and Funding Arrangement

In terms of construction scale, this project is not a single-function webpage but a platform-oriented software system for university course scenarios. The current system has already formed a relatively complete course-level AI teaching loop, including course management, material ingestion and RAG retrieval, chat-based question answering, assignment submission and AI grading, learning analytics, personalized exercise generation, platform embedding, and Agent Builder. Both the front-end and the back-end already have relatively complete demonstration capability.

From the perspective of technical implementation, the project adopts a front-end/back-end separation architecture with multi-service collaboration. The front end uses technologies such as React, TypeScript, and Vite, while the back end uses FastAPI, PostgreSQL, Redis, Celery, and MinIO, and integrates large-model interfaces for question answering, grading, and document processing. This indicates a relatively high level of technical complexity, and the project cost is reflected not only in the up-front software development investment but also in later server operation, model invocation, and operation-maintenance management expenses. Given that this experiment uses a simplified economic evaluation scope, the subsequent cost estimation mainly reflects this technical complexity through development cost, operation-maintenance labor cost, and large-model API cost.

According to the experimental requirement, the analysis first proceeds under the assumption of no debt financing, that is, the project is initially assumed to be funded by self-owned capital so that the profitability and investment feasibility of the project itself can be analyzed. As for the evaluation model, this report adopts a cash flow model and prepares a series of supporting financial analysis statements, including the investment estimate table, working capital estimate table, operating revenue table, VAT and surtax estimate table, fixed asset depreciation table, intangible asset amortization table, and total cost estimate table, and then further forms the project investment cash flow statement for the subsequent economic evaluation.

3.2 Supporting Financial Analysis Tables

The initial investment of this project mainly consists of software development cost and initial deployment cost. The software development cost is taken from the result of the previous software size and cost estimation experiment. According to the previous function point analysis, the adjusted function point count is about 228, corresponding to approximately 8.82 person-months of development workload. Based on a labor rate of RMB 31,309 per person-month, the software development cost is estimated at RMB 276,145.38. Since testing-related labor may already be included in the development cost, this report does not count testing labor separately again so as to avoid double counting. In addition, the estimate also considers initial deployment, environment configuration, and other initial preparation expenses.

Table 1 Investment Estimate Table

Item	Amount	Description
Software development cost	276145.38	Estimated from previous function point analysis and labor rate
Initial deployment and configuration cost	2000.00	For first-time deployment and environment configuration
Other initial preparation expenses	2000.00	For documentation, initialization, and related preparation
Total construction investment	280145.38	Total input during the construction stage

Working capital is mainly used for daily turnover during the initial operating stage of the project. Based on the actual deployment situation, the server is a cloud server rented at RMB 500 per month, and the database is deployed by Docker on the same server, so the database is not charged separately. Considering that the project needs to retain a certain amount of working capital at the start-up stage, this report estimates working capital based on three months of operating expenditure.

Table 2 Working Capital Estimate Table

Item	Amount (RMB)	Description
Working capital	9000.00	Estimated based on three months of operating expenses

Considering that university platform projects usually follow a promotion path of 'pilot first, expansion later', it is assumed that the project serves 6 courses and completes 1 customized configuration in Year 1, serves 12 courses and completes 2 customized configurations in Year 2, and serves 22 courses and completes 4 customized configurations in Year 3.

Table 3 Operating Revenue Table

Item	Year 1	Year 2	Year 3
Number of courses under annual course-level service	6	12	22
Number of customized configurations	1	2	4
Revenue from annual course-level service (RMB)	60000	120000	220000
Revenue from customized configuration service (RMB)	12000	24000	48000
Total operating revenue (RMB)	72000	144000	268000

This project belongs to the category of software platforms and educational information services. Therefore, this report estimates VAT at a rate of 6%; urban maintenance and construction tax is taken as 7% of VAT; education surcharge is taken as 3% of VAT; and local education surcharge is taken as 2% of VAT.

Table 4 VAT and Surtax Estimate Table

Item	Year 1 (RMB)	Year 2 (RMB)	Year 3 (RMB)
Operating revenue	72000	144000	268000
VAT (6%)	4320	8640	16080
Urban maintenance and construction tax	302.4	604.8	1125.6
Education surcharge	129.6	259.2	482.4
Local education surcharge	86.4	172.8	321.6
Total taxes and surcharges	518.4	1036.8	1929.6

Since the project is mainly based on software platforms and cloud resources, and does not involve large-scale procurement of dedicated hardware, the proportion of fixed assets is relatively small. It is assumed that a small amount of fixed assets is formed during deployment, with an original value of RMB 5,000, a residual value rate of 5%, and a depreciation period of 5 years. Therefore, the annual depreciation is RMB 950.

Table 5 Fixed Asset Depreciation Table

Item	Value
Original value of fixed assets (RMB)	5000
Residual value rate	5%
Depreciation period (years)	5
Annual depreciation (RMB)	950

The core value of the project is mainly embodied in the software platform itself, so part of the platform software achievements can be treated in a simplified way as intangible assets. Assuming an original value of RMB 10,000 and an amortization period of 5 years, the annual amortization is RMB 2,000.

Table 6 Intangible Asset Amortization Table

Item	Value
Original value of intangible assets (RMB)	10000
Amortization period (years)	5
Annual amortization (RMB)	2000

Considering the operating characteristics of the project, the main costs during the operating stage include server cost, large-model API cost, operation-maintenance labor cost, fixed asset depreciation, intangible asset amortization, and taxes and surcharges. The server expense uses the actual rental price of the project, namely RMB 500 per month and RMB 6,000 per year. The database shares the same deployment environment with the server and is therefore not charged separately. The operation-maintenance labor cost is estimated based on benchmark data and is temporarily treated as a fixed annual cost. The model invocation cost increases year by year as the number of covered courses expands.

Table 7 Total Cost Estimate Table

Item	Year 1 (RMB)	Year 2 (RMB)	Year 3 (RMB)
Cloud server	6000	6000	6000
Database	0	0	0
Large-model API calls	5000	10000	20000
Operation-maintenance labor cost	24495.43	24495.43	24495.43
Depreciation	950	950	950
Intangible asset amortization	2000	2000	2000
Taxes and surcharges	518.4	1036.8	1929.6
Total cost	38963.83	44482.23	55375.03

For clearer presentation in the cash flow model format, the previous investment, cost, and revenue estimates are consolidated into the following summary table. Detailed calculations remain subject to the supporting financial tables and the formal project investment cash flow statement.

Table 7-1 Comprehensive Cash Flow Model Summary Table

Item	Year 0 (RMB)	Year 1 (RMB)	Year 2 (RMB)	Year 3 (RMB)
I. Initial Investment				
Software development cost	276145.38	0	0	0
Initial deployment and configuration cost	2000.00	0	0	0
Other initial preparation expenses	2000.00	0	0	0
Working capital	9000.00	0	0	0
Total initial investment	289145.38	0	0	0
II. Ongoing Costs				
Cloud server	0	6000	6000	6000
Database	0	0	0	0
Large-model API calls	0	5000	10000	20000
Operation-maintenance labor cost	0	24495.43	24495.43	24495.43
Depreciation	0	950	950	950
Intangible asset amortization	0	2000	2000	2000
Taxes and surcharges	0	518.40	1036.80	1929.60
Total ongoing costs	0	38963.83	44482.23	55375.03
III. Revenue				
Annual course-level service revenue	0	60000	120000	220000
Customized configuration service	0	12000	24000	48000

revenue				
Total revenue	0	72000	144000	268000
IV. Annual net operating cash flow	-289145.38	33036.17	99517.77	212624.97
Recovery of working capital in Year 3	0	0	0	9000.00
V. Adjusted net cash flow	-289145.38	33036.17	99517.77	221624.97

3.3 Pre-Financing Analysis

According to the experimental requirement, the project is first analyzed before financing, that is, without considering debt financing, and only from the perspectives of project investment and operation. The cash outflow in Year 0 includes construction investment and working capital. The net cash flow from Year 1 to Year 3 equals operating revenue minus total cost. At the end of Year 3, the full working capital of RMB 9,000 is recovered.

Table 8 Project Investment Cash Flow Statement

Item	Year 0	Year 1	Year 2	Year 3
Cash inflow	0	72000	144000	277000
Cash outflow	289145.38	38963.83	44482.23	55375.03
Net cash flow	-289145.38	33036.17	99517.77	221624.97

Net present value reflects the net earnings level of the project after considering the time value of money, and its formula is $NPV = \Sigma[NCF_t/(1+i)^t]$, where NCF_t is the net cash flow in year t and i is the discount rate, which is uniformly set at 8% in this experiment. The internal rate of return is the discount rate that makes the project NPV equal to zero and can be calculated using the IRR function in Excel. The dynamic payback period is the time required for the cumulative discounted cash flow to turn from negative to positive after considering the time value of money.

Table 9 Economic Evaluation Indicator Results

Indicator	Result	Description
Net present value	2697.16 RMB	Greater than 0, indicating a certain net earning capability under the current assumptions
Internal rate of return	About 8.40%	Slightly higher than the benchmark rate of return of 8%
Dynamic payback period	About 2.98 years	Basic investment recovery is completed within the 3-year evaluation period

3.3.4 Conclusion of the Pre-Financing Analysis

Based on the project investment cash flow statement and the calculated economic evaluation indicators, the project can be considered to have a certain degree of pre-financing economic feasibility under the current assumptions. The NPV is RMB 2,697.16, which is greater than 0; the IRR is about 8.40%, slightly higher than the benchmark rate of return of 8% adopted in this experiment; and the DPP is about 2.98 years, indicating that the project can basically recover its investment within the 3-year evaluation period.

At the same time, however, the economic performance of the project is still relatively limited. The main reason is that the up-front software development investment is high, while the course coverage and revenue scale in the initial operating stage are still limited. Therefore, the economic feasibility of the project depends to a large extent on the continuous increase in the number of courses, the improvement of platform reuse efficiency, and the reasonable control of model invocation cost and operation-maintenance cost. Overall, the project is not a short-term high-return project under the current assumptions, but it already has certain medium- and long-term promotion value and economic feasibility after further optimization.

3.3.5 Result Verification and Reasonableness Analysis

To ensure the reasonableness of the economic evaluation results, this report further verifies the main parameters and calculation results. First, with respect to development cost, the software development cost uses the result of the previous software size and cost estimation experiment and is calculated on the basis of function point scale, productivity, and person-month labor rate, which gives it support from previous experimental data. At the same time, testing-related labor has not been counted again in this experiment, thereby avoiding double counting.

Second, with respect to operating cost, the server expense uses the actual rental price of the project, and the database is not charged separately because it shares the same deployment environment with the server. The operation-maintenance labor cost is estimated based on the operation-maintenance productivity and the Shanghai software operation-maintenance labor rate in the 2025 China Software Industry Benchmark Data. The large-model API cost is budgeted according to expected usage and the official token charging rules and is treated as a budget item for the operating stage.

In terms of revenue assumptions, this report does not directly regard the project as a mature institution-level teaching platform such as Xuexitong or Zhihuishu. Instead, based on its current positioning as a course-level AI teaching platform prototype, the report adopts a revenue model of annual course-level service fees plus customized configuration service fees, and assumes that course coverage expands year by year along a path of pilot first and expansion later. Therefore, although the revenue estimate is predictive to a certain extent, it is still generally consistent with the current development stage of the project.

Judging from the calculated results, the project has a positive NPV, an IRR slightly higher than the benchmark rate of return, and a DPP close to 3 years. This indicates that its economic performance is not overly generous, but it does have a certain degree of feasibility. This result is basically consistent with the actual characteristics of the project, namely high up-front development investment and later reliance on expansion in course scale and improvement in platform reuse to increase returns. Therefore, the economic evaluation results in this report can be regarded as reasonably valid and useful for reference under the current assumptions.

3.4 Post-Financing Analysis

For the post-financing analysis, this report adopts a simplified financing assumption in which 60% of the total project investment is funded by self-owned capital and the remaining 40% is supported by external financing. The total project investment is RMB 289,145.38, including RMB 173,487.23 of self-owned capital and RMB 115,658.15 of external financing. The financing term is assumed to be 3 years, the annual interest rate is set at 4%, and the repayment method is equal principal repayment. This assumption is introduced only for analytical purposes in order to examine the return on equity capital under reasonable financing conditions and does not imply that the project has already secured actual external financing.

Under this financing scheme, the annual principal repayment is RMB 38,552.72. Interest is calculated based on the outstanding financing balance at the beginning of each year, resulting in interest payments of RMB 4,626.33 in Year 1, RMB 3,084.22 in Year 2, and RMB 1,542.11 in Year 3. According to the pre-financing cash flow statement, the project net cash flows are RMB 33,036.17, RMB 99,517.77, and RMB 221,624.97 for Years 1 to 3, respectively. On this basis, the post-financing equity capital cash flow statement can be prepared.

Table 10 Post-Financing Equity Capital Cash Flow Statement

Item	Year 0	Year 1	Year 2	Year 3
Self-owned capital input	-173487.23	0	0	0
Project net cash flow	0	33036.17	99517.77	221624.97
Principal repayment	0	38552.72	38552.72	38552.72
Interest payment	0	4626.33	3084.22	1542.11
Equity capital net cash flow	-173487.23	-10142.88	57880.83	181530.14

Table 11 Post-Financing Economic Evaluation Indicator Results

Indicator	Result	Description
Equity capital NPV	9696.22 RMB	Greater than 0, indicating that the equity capital still has a certain earning capability after financing

Equity capital IRR	About 10.14%	Higher than the benchmark rate of return of 8%
Dynamic payback period	About 2.96 years	Basic investment recovery is completed within the 3-year evaluation period

The post-financing analysis shows that after introducing a certain proportion of external financing support, the IRR on equity capital becomes higher than the project IRR in the pre-financing analysis, indicating that reasonable financing improves the efficiency of capital use to some extent. Meanwhile, the equity capital NPV remains positive and the DPP stays within 3 years, which suggests that the project still has a certain degree of economic feasibility under the current financing scheme.

However, it should also be noted that the post-financing analysis is likewise based on the assumptions that the course scale continues to expand and that the annual course-level service fees and customized configuration service fees can be realized as expected. If the actual promotion scale is lower than expected, or if model invocation cost and operation-maintenance cost exceed the budget, the financial performance after financing will also be affected. Therefore, the post-financing feasibility of the project still depends on later promotion effects and cost control.

In this post-financing analysis, the project investment IRR is higher than the assumed financing cost, so moderate external financing support produces a certain leverage effect and results in an equity capital IRR that is higher than the pre-financing project IRR. This outcome is financially reasonable.

4. Experimental Results and Conclusion

By combining the results of the pre-financing and post-financing analyses, the project can be considered to have a certain degree of economic feasibility under the current assumptions. The pre-financing analysis shows an NPV of RMB 2697.16, an IRR of 8.40%, and a DPP of 2.98 years. The post-financing analysis shows an equity capital NPV of RMB 9696.22, an equity capital IRR of 10.14%, and a DPP of 2.96 years. This indicates that as long as the course scale can gradually expand and the platform reuse efficiency can continue to improve, the project still has a certain promotion potential in the medium and long term.

Nevertheless, this conclusion does not mean that the project already has a clear short-term profitability advantage. Because the up-front development investment is large while the revenue scale in the initial operating stage is relatively limited, the actual return of the project still depends heavily on the number of connected courses, the control of model invocation cost, the improvement of platform stability, and the promotion effect in real teaching scenarios. Therefore, subsequent work should continue to optimize the cost structure, enhance the general capability of the platform, improve the efficiency of course access, and gradually refine the operation and service system.

5. Reflection

Through this experiment, I gained a clearer understanding of the complete process of software project economic evaluation. In the past, when working on course projects, I paid more attention to whether functions were implemented and whether the system could run. This experiment further made me realize that whether a project is worth promoting also needs to be judged comprehensively from perspectives such as market demand, cost structure, charging model, and cash flow performance.

During the writing of the report, I developed a deeper understanding of the relationship among function points, development cost, person-month labor rate, operation-maintenance productivity, tax parameters, and cash flow indicators. I also became more aware of the foundational role of supporting financial analysis tables in project evaluation. In particular, distinguishing between project investment cash flow and equity capital cash flow under the pre-financing and post-financing evaluation scopes gave me a more concrete understanding of the meanings of NPV, IRR, and DPP.

Overall, this experiment not only helped me master the basic logic of software project economic evaluation, but also improved my ability to analyze software projects from the perspectives of engineering management and economic decision-making. This will be useful for further refining the course project, participating in project defense, and continuing the study of software engineering management and economics.

6. Thinking Question

If a software project or product has no direct cash income, its benefits cannot be evaluated only by cash-flow-based financial indicators such as NPV, IRR, and DPP. The reason is that such indicators presuppose that the project can generate measurable monetary returns, whereas projects without direct cash income often emphasize efficiency improvement, service optimization, social value, or long-term strategic significance.

For such projects, a comprehensive benefit evaluation method is more appropriate. Specifically, the analysis may focus on several aspects: whether the project improves work efficiency, such as reducing teachers' repetitive labor or shortening the time for material organization and question answering; whether it improves service quality, such as enhancing teaching support capability, feedback timeliness, and personalized tutoring; whether it has social benefits or public value, such as aligning with the direction of educational digitalization and smart education construction; and whether it has long-term strategic significance, such as providing a basis for later platform reuse, course expansion, and data accumulation.

Therefore, for software projects without direct cash income, the evaluation of benefits should shift from pure financial profitability analysis to comprehensive benefit analysis. When necessary, cost-effectiveness analysis, analytic hierarchy process, or multi-indicator comprehensive evaluation can be combined to systematically judge the project's benefits in terms of efficiency, quality, social value, and long-term application significance.

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